Prob. 4: Consider the circuit shown where $V_x = 20V$ and $I_s = 4A$. Using ***mesh method***.

a. Write all the required equations to find i_1, i_2, i_3, and v_o (***Do not simplify***)

b. Solve to find i_1, i_2, i_3, and v_o.

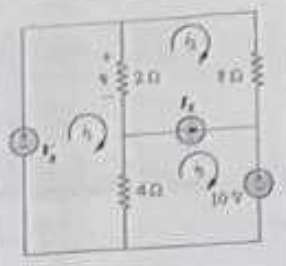


Figure 4

a.

$$8I_2 + 10 + 4(I_3 - I_1) - 2(I_2 - I_1) = 0$$

$$-2I_1 + 6I_2 + 4I_3 = -10 \Rightarrow (1)$$

$$I_3 + I_s = I_2$$

$$I_3 + 4 = I_2$$

$$I_2 - I_3 = 4 \Rightarrow (2)$$

$$2(I_1 - I_2) + 4(I_1 - I_3) = -V_x$$

$$6I_1 - 2I_2 - 4I_3 = -20 \Rightarrow (3)$$

b.

$i_1 = -6.75A$; $i_2 = -0.75A$

$i_3 = -4.75A$; $v_o = 2(I_1 - I_2) = -12V$

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Prob. 1: Consider the circuit shown where $V_s = 30V$.

a. Using voltage divider, find V_x and V_{ab}.

b. Using Ohm's law, find I_s and I.

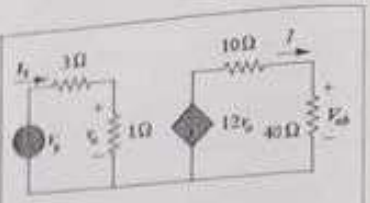


Figure 1

a. V_x = ... 7.5V . . . . . V_{ab} = ... -72V . . . . . (6)

b. $I_s = \frac{V_s}{R_{eq}} = \frac{30}{4} = 7.5A$

$I = \frac{-12V_x}{R_{eq}} = \frac{-90}{50} = -1.8A$

Prob. 2: Consider the circuit shown where $R = 14\ \Omega$.

a. Find the equivalent resistance between the a-b terminals.

b. When the voltage between the *a-b* terminals is 84 V, find the current I and the voltage V_x.

c. When $I = 10A$, using current division, find the current through the 30-Ω resistance.

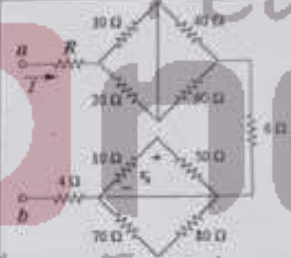


Figure 2

a. R_{eq} = . 61.33Ω . .

b. $I = \frac{V}{R_{eq}} = \frac{84}{61.33} = 1.367A$

V_x = . 0V . (short circuit)

c. $I_{30} = \frac{V}{R_{3\Omega}} = \frac{84}{30} = 2.8A$

Prob.3:

(a) The charge transferred through a circuit element is givin by $q(t)=6t^2+6t$ C Calculate the current entering the element at instant $t = 10ms$.

$i(t)$ = . 126A . . . . . .

(b) If a load draws 1kW when connected to a 230 V source, then the resistance of the load (R_L) is

R_L = 15.16Ω . .

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Prob. 5: Consider the circuit shown where $I_s = 2A$. Using **nodal method**:

a. Write **all** the required equations to find v_1, v_2, v_3, and I_o (*Do not simplify*)

b. Solve **to** find v_1, v_2, v_3, and I_o.

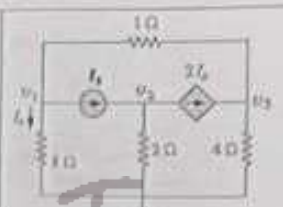


Figure 5

a

$$I_o + I_s + \frac{v_1 - v_3}{1} = 0 \quad \ldots (1)$$

$$I_o + v_1 - v_3 = -2$$

$$\frac{-v_1}{8} - v_3 = -2 \Rightarrow (1)$$

$$-I_s + \frac{v_2}{2} + 2I_o = 0$$

$$2I_o + \frac{v_2}{2} = 2$$

$$\frac{7v_1}{4} + \frac{v_2}{2} = 2 \Rightarrow (2)$$

$$\frac{v_3}{4} + \frac{v_3 - v_1}{1} - 2I_o = 0$$

$$-v_1 + \frac{5v_3}{4} - 2I_o = 0$$

$$\frac{-11v_1}{4} + \frac{5v_3}{4} = 0 \Rightarrow (3)$$

$I_o = \frac{v_1}{8}$

b.

$v_1 = 1.5\,V$; $v_2 = -1.28V$

$v_3 = 3.32V$; $I_o = \frac{1.5}{8} = 0.1875A$

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